

Sharif University of Technology | Department of Computer Engineering
MSc Thesis Defense in Computer Engineering: Artificial Intelligence and Robotics

Predicting Catastrophic Overfitting in Fast Adversarial Training

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Defense Roadmap: From Failure to Step-Size Control

1. Problem: why can high single-step accuracy be misleading?
2. Analysis and method: loss geometry, PertAlign, and the SORA controller.
3. Evidence: robustness, transfer of settings, computational cost, and limits.

Scientific basis: the joint SORA paper at ICML 2026 [1]; thesis Chapters 1–7 and the appendix.

Practical Question and Threat Model

- Detect and correct a failing single-step approximation using information already available during training.
- An untargeted, white-box attacker increases the image-classification loss.
- The metric is empirical accuracy under strong attacks, not a mathematical robustness certificate.

$$\Delta_\infty(x, \epsilon) = \{\delta : \|\delta\|_\infty \leq \epsilon, \quad x + \delta \in [0, 1]^d\}$$

Threat model and evaluation: [2, 3].

Adversarial Training: Two Coupled Problems

$$\min_{\theta} \; \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\delta \in \Delta_\infty(x,\epsilon)} \mathcal{L}(f_\theta(x+\delta), y) \right]$$

- Inner problem: find a high-loss example for the current model.
- Outer problem: update the weights using that example.
- Fast training reduces the cost of the inner solve; its approximation quality needs monitoring.

Formulation: [2]; fast training: [4].

One Gradient Versus Repeated Direction Updates

Fast Gradient Sign Method (FGSM): a first-order approximation

$$g = \nabla_x \ell(x), \qquad x_{\text{FGSM}} = \Pi_{[0,1]^d}(x + \epsilon \operatorname{sign}(g))$$

Projected Gradient Descent (PGD): a multi-step attack

$$\delta^{t+1} = \Pi_{\Delta_\infty(x,\epsilon)} \left(\delta^t + a \operatorname{sign}\big(\nabla_x \ell(x + \delta^t)\big)\right)$$

- FGSM keeps the initial direction; PGD observes changes in direction.
- The perturbation budget ϵ and step size a are different quantities.

FGSM: [5]; PGD: [2].

Catastrophic Overfitting: Apparent Solver Success

- Accuracy under a multi-step attack collapses over a short interval.
- Accuracy under the single-step training attack may remain high.
- Low loss at the attack endpoint does not bound the maximum loss nearby.

Takeaway

Assess the training-example generator itself: evaluation with the training attack alone is insufficient.

Definition and analysis: [4, 6].

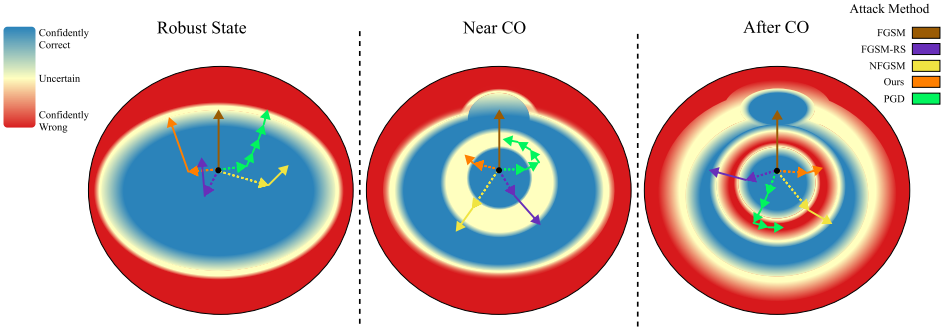
What Do Previous Methods Change?

Family	Examples	Mechanism and open question
Perturbation diversity	FGSM-RS, N-FGSM	Vary starts and amplitudes; do settings transfer?
Regularization	GradAlign, NuAT, ELLE	Control geometry; at what computational cost?
Abnormal examples	AAER	Control loss decrease; how early is the signal?
Adaptive steps	ATAS	Use gradient magnitude; what about curvature?

New synthesis of the review in SORA Section 2 [1]; this table does not appear in the paper or seminar.

Methods: [7, 6, 8, 9, 10, 11].

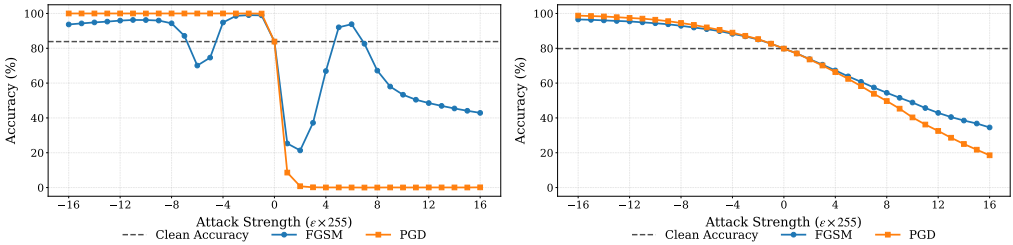
Geometry: One Step Can Skip a Vulnerable Region



Decision-boundary distortion and loss-surface evolution; reproduced from SORA Figure 1 [1].

A low-loss endpoint can hide high-loss points along the path.

Epsilon Overfitting: Sensitivity to Perturbation Size



(a) FGSM Adversarial Training

(b) SORA Adversarial Training

Accuracy across attack sizes; reproduced from SORA Figure 2 [1].

- Learning at particular sizes does not protect the whole neighborhood.
- Negative axis values reverse the direction; the threat radius is not negative.

Abnormal Adversarial Examples: Overshooting a Peak

$$z = x + \eta, \qquad \ell(z + \delta) < \ell(z)$$

- The attack aims to increase loss, yet reduces it at the endpoint.
- An AAE is abnormal in terms of loss, not necessarily label correctness.
- The gradient direction can change before this loss decrease appears.

Takeaway

Comparing two gradients can provide information earlier than comparing two losses.

AAE definition: [10]; motivation for geometric monitoring: [1].

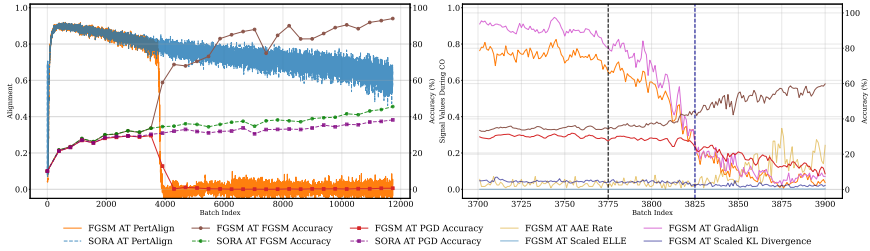
Perturbation Alignment: Two Available Gradients

$$g = \nabla \ell(z), \quad g' = \nabla \ell(z + \delta), \qquad P = \frac{g^{\mathrm{T}} g'}{\|g\|_2 \|g'\|_2}$$

- PertAlign compares gradient directions at the start and end of the attack.
- The first gradient constructs the attack; the second accompanies backpropagation on the adversarial example.
- A value near one means directional agreement, not necessarily low curvature.

Equation 2, p. 5 of SORA [1]; defined for nonzero gradients.

Warning Evidence: About 50 Batches



Alignment falls before accuracies separate; reproduced from SORA Figure 3 [1].

- One FGSM trajectory: decline around batch 3775; clear separation around 3825.
- This is not an average or guaranteed lead time; no threshold or false-alarm rate is reported.

Geometric Relation: Projected Hessian Action

For $g \neq 0$, a fixed direction v, and a small step α:

$$H = \nabla^2 \ell(z), \qquad Q_g = I - \frac{gg^{\mathrm{T}}}{\|g\|_2^2}$$

$$1 - P = \frac{\alpha^2}{2\|g\|_2^2} \|Q_g H v\|_2^2 + O(\alpha^3)$$

- Assumptions: local smoothness and a locally Lipschitz Hessian; hold the direction fixed in the expansion.
- $P \approx 1$ does not imply a small $\|H\|$: parallel gradient changes are invisible.

Projected reformulation in the thesis geometry chapter, based on Appendix A of SORA [1].

Exact Theory: Scalar Optimum of a Quadratic Model

$$p = \operatorname{sign}(g), \quad q(a) = \ell(z) + ba + \tfrac{1}{2}ca^2, \quad b = \|g\|_1 > 0,\ c = p^{\mathrm{T}}Hp$$

$$\arg\max_{0 \le a \le A} q(a) = \begin{cases} \min\{A, -b/c\}, & c < 0, \\ A, & c \ge 0. \end{cases}$$

- In a concave direction, moving beyond the vertex reduces loss.
- This solves the scalar model exactly; the real loss need not be quadratic.

Thesis method derivation, based on the second-order analysis in SORA [1].

Curvature Information from the Same Probe Step

In a quadratic model, without clipping, for a scalar probe $a_{\text{probe}} > 0$:

$$g_{\text{probe}} = g + a_{\text{probe}} Hp, \qquad r = \frac{p^{\mathrm{T}} g_{\text{probe}}}{\|g\|_1} = 1 + \frac{a_{\text{probe}} c}{b}$$

$$r < 1 \quad \Longrightarrow \quad a_{\text{quad}} = \min\left\{ A, \frac{a_{\text{probe}}}{1 - r} \right\}$$

- The numerator is the actual step used to measure the second gradient.
- The ratio r is not the PertAlign cosine: gradient magnitude also matters.

Exact reformulation in the thesis method chapter from the SORA derivation [1].

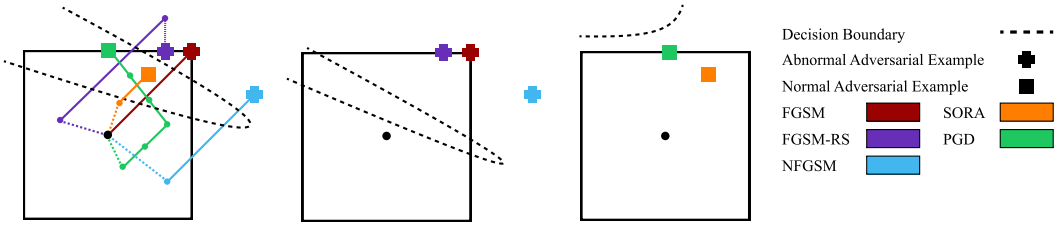
Practical Rule: Quadratic-Inspired Amplitude Control

$$F(u) = \begin{cases} \min\left\{\alpha_{\max}, \dfrac{\alpha_0}{1-u}\right\}, & u < 1, \\ \alpha_{\max}, & u \geq 1, \end{cases} \qquad a_j \sim \mathcal{U}(0, A_i)$$

- u is an exponential moving average of the ratio; A_i caps the componentwise step distribution.
- Replacing the probe step by α_0 changes the previous exact identity.
- Componentwise randomness and use of past batches are further practical choices.

SORA Algorithm 1 [1]: $\alpha_0 = 0.02$, $\beta = 0.05$, $\alpha_{\max} = 2\epsilon$.

Why Can Step-Size Correction Help?

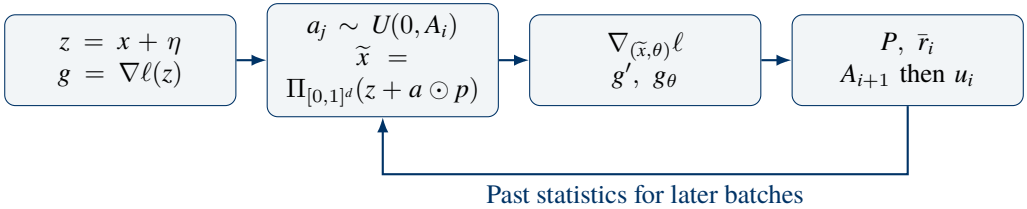


(a) Before update (b) After update on AAEs (c) After update on NAEs

Geometric interpretation of step-size correction; reproduced from SORA Figure 4 [1].

Reaching high-loss points can redirect training toward the vulnerable region.

The SORA Pipeline: Reusing Two Passes



New diagram of Algorithm 1 [1]; this diagram does not appear in the original paper or seminar.

- The second pass yields both the weight derivative and endpoint input gradient.
- PertAlign monitors training; the ratio r controls the step cap.

Pseudocode and the Printed Order's Extra Delay

1. Initialize $u_0 = 0.99$, $A_1 = F(u_0)$; for each batch draw $\eta \sim U([-\epsilon, \epsilon]^d)$.
2. $g = \nabla \ell(x + \eta)$, $p = \text{sign}(g)$; detach the attack direction.
3. Draw $a_j \sim U(0, A_i)$; form $\widetilde{x} = \Pi_{[0,1]^d}(x + \eta + a \odot p)$.
4. Obtain (g', g_θ); update weights; compute $\bar{r}_i$ by averaging per-example ratios.

$$\boxed{A_{i+1} = F(u_{i-1})} \quad \text{then} \quad u_i = (1 - \beta)u_{i-1} + \beta \bar{r}_i$$

Batch i's observation can first affect the cap at batch $i + 2$.

Algorithm 1, p. 6 [1]; A_1 initialization and mean of ratios are explicit thesis method conventions.

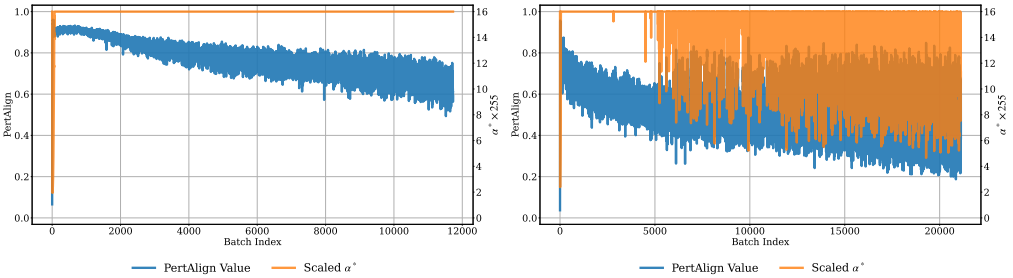
Training Domain and Evaluation Threat Ball Differ

$$\widetilde{x} = \Pi_{[0,1]^d}(x + \eta + a \odot p), \qquad \|\eta + a \odot p\|_\infty \leq \epsilon + \alpha_{\max} = 3\epsilon$$

- Training: clip pixels to $[0, 1]$ only; no projection onto the ϵ-ball.
- 3ϵ bounds the raw perturbation; it is not every example's fixed size.
- Evaluation: the ϵ-ball and pixel domain; usually $8/255$, but $3/255$ for TissueMNIST.

Algorithm 1 and Appendix D, p. 31 of SORA [1].

Adaptive Behavior Differs Across Two Datasets



(a) CIFAR-10 (b) PATHMNIST

Alignment and step cap; reproduced from SORA Figure 5 [1].

The cap is often active on CIFAR-10; step reductions are more frequent on PathMNIST.

Training Settings and Evaluation Scope

- Datasets: CIFAR-10/100, TinyImageNet, ImageNet-100, PathMNIST, and TissueMNIST.
- Architectures: PreActResNet-18, ResNet-18, WRN-28-10, and SENet-18; models are trained separately.
- SGD: momentum 0.9, weight decay 5×10^{-4}; cyclic learning rate for natural images, cosine for medical images.

Takeaway

"Fixed settings" means a shared final parameter set after limited tuning; data preparation and all training details are not identical.

Section 6 and Appendices D, E, M.3, pp. 30–34 and 60 of SORA [1].

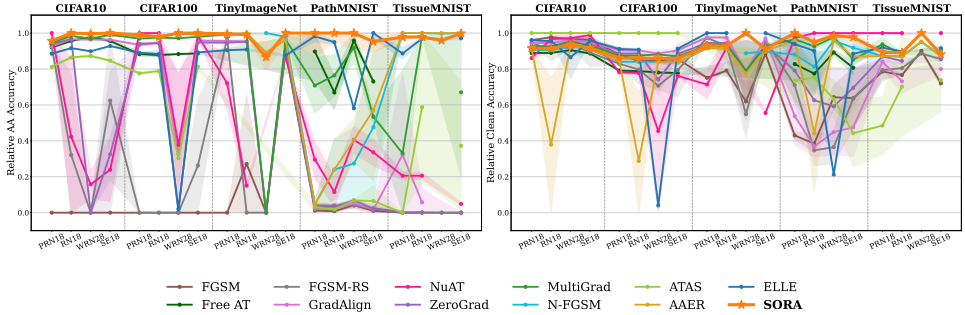
Main Results on Two Difficult Problems

	PathMNIST		ImageNet-100	
Method	Clean	AA	Clean	AA
FGSM	46.06	0.42	15.94	0.00
N-FGSM	74.89	1.65	49.40	15.52
AAER	83.54	2.48	48.26	17.18
SORA	**85.93**	**32.91**	**57.26**	**18.56**

Selected data from Table 1, p. 9 [1]; PreActResNet-18, $\epsilon = 8/255$; all values are percentages.

AA gains over N-FGSM: 31.26 and 3.04 percentage points, respectively.

Transfer of Settings: Relative Performance Overview



Reproduced from Figure 6, p. 8 [1]; normalized by each problem's best value, as described on p. 47.

- A value near one means proximity to the best result in that column.
- This view hides absolute values and the size of small differences.

Detailed Tables: No Method Wins Every Column

Dataset / model	Method	Clean	PGD-10	AA
CIFAR-10 / PreAct-18	N-FGSM	79.05	48.73	43.95
	SORA	79.90	48.64	43.83
CIFAR-100 / PreAct-18	N-FGSM	53.43	26.25	21.75
	SORA	54.74	26.12	21.56
TinyImageNet / WRN-28-10	N-FGSM	48.47	21.21	**16.58**
	SORA	54.59	18.73	14.31

New arrangement from Tables 20, 24, and 30, pp. 48, 50, and 53 [1]; central percentage values only, omitting $\pm$; not an original paper table.

Interpretation: a clean–robust–cost tradeoff; rankings depend on the problem.

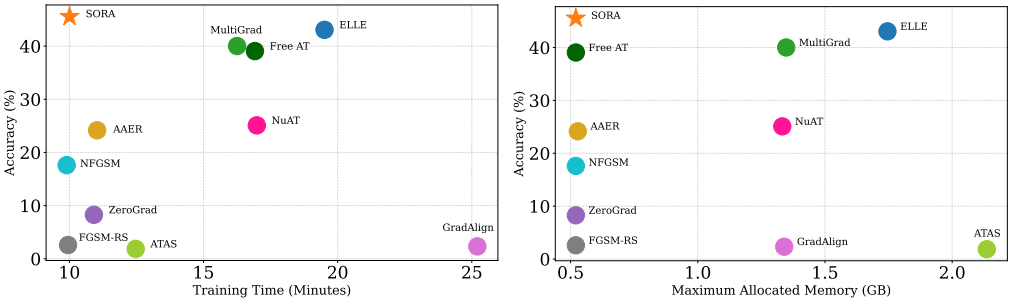
Component Ablation: A Separate PGD-10 Report

Setting	Clean	FGSM	PGD-10
SORA (baseline)	84.69	57.51	45.56
Without random sampling	85.67	58.89	45.35
Clamping step size	86.82	52.02	34.08
Without adaptive step size	89.93	28.30	17.23

Table 2, p. 9 [1]; PathMNIST/PreActResNet-18, percentages; independent of the main AA report.

Removing adaptation causes a large drop; the 0.21-point sampling drop alone does not establish statistical significance.

Efficiency: Time and Memory in One Setting



(a) Time (b) Memory

Reproduced from Figure 7, p. 8 [1]; PathMNIST/PreActResNet-18, 30 epochs, RTX 4090; accuracy measured with PGD-10.

Evidence Limits and Defensible Claims

- Warning: an observed temporal lead, without a calibrated predictor or false-alarm rate.
- Theory: local and direction dependent; randomization, clipping, and delay depart from the exact model.
- Evaluation: ImageNet-100 has one run; full attack details and relationships between some reports remain unclear.

Thesis results, conclusions, and technical appendix; SORA Appendices J and M [1].

Contributions: Connecting Geometry to Training Control

1. Epsilon-overfitting interpretation: cover diverse perturbation sizes and directions.
2. PertAlign: monitor gradient-direction changes using existing computation, with a local geometric interpretation.
3. SORA: control the step-distribution cap using two gradients; a competitive tradeoff within the evaluated scope.

Contributions of the joint research [1]; this defense presents the documented narrative and analysis of the completed thesis chapters.

Future Work: Toward a Measurable Predictor

- Define collapse independently, choose a validation threshold, and measure false alarms and lead time across multiple runs.
- Study per-example or group control for imbalanced datasets; weigh gains against variance and cost.
- Separate amplitude, diversity, and adaptation under matched computational budgets and training domains.

Takeaway

Two existing gradients can support both warning and step-size correction; independent robustness evaluation remains essential.

Directions proposed in the thesis conclusions; limitations of the joint study [1].

Backup: Report Provenance and an Unresolved Difference

- Main Table 1, p. 9: PathMNIST, clean 85.93 and AA 32.91.
- Ablation Table 2, p. 9: clean 84.69 and PGD-10 45.56; a separate metric.
- Table 32, p. 53: clean 84.88 ± 0.99 and AA 35.54 ± 1.55; a separate appendix report.

All concern PathMNIST/PreActResNet-18 in SORA [1]; the exact relationship between runs is unclear. Values have not been merged or averaged.

Backup: Reproduction and Structural Cost

Method	Forward passes	Backward passes
FGSM-RS / SORA	2	2
GradAlign / ELLE	3	2
PGD-2	3	3
PGD-10	11	11
TRADES	12	11

New selection and arrangement from Table 19, p. 48 [1]; counts per batch; not the complete original table.

- Record seeds, checkpoints, AA version and settings, attack steps, and restarts.
- Specify ratio reduction, zero-denominator handling, initial cap, and EMA order.

Terminology: Problem and Evaluation

Term	Meaning in this presentation
Adversarial	Chosen to increase the classifier's loss.
Perturbation	An additive input change δ.
Dataset	The collection of labeled examples for a task.
Accuracy	The fraction of examples classified correctly.
Robustness	Performance under the specified allowed perturbations.
CO	Catastrophic overfitting: rapid multi-step accuracy collapse despite high single-step accuracy.

New terminology guide based on the thesis glossary and paper terminology [1]; not an original paper table.

Terminology: Geometry and Method

Term	Meaning in this presentation
Gradient	First derivative of the loss with respect to the input.
Curvature	Local gradient variation described by Hessian action.
Loss	The classification objective $\ell(z) = \mathcal{L}(f_\theta(z), y)$.
PertAlign	Perturbation Alignment: cosine of start and endpoint gradients.
AAE	Abnormal Adversarial Example: an attack endpoint with lower loss than its reference start.
EMA	Exponential Moving Average: here, β weights the new observation.

New guide to the paper's terms [1] and the thesis glossary; not an original paper table.

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